

ALICIA JUANITA LISAL

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SUMMARY

Informatics student specializing in Artificial Intelligence, with hands-on experience in machine learning, data-driven applications, and full-stack development. Built ML pipelines for real-world problems using Python, scikit-learn, and PySpark, including a published vaccine demand forecasting study and a YOLOv11-based computer vision system. Strong interest in model development, data preprocessing, and scalable AI system design.

SKILLS

ML / Data: Python, scikit-learn, Pandas, NumPy, PySpark, Ultralytics (YOLOv11), Matplotlib, Jupyter Notebook

Languages: Python, PHP, Dart, C#, SQL

Frameworks: Flutter, Laravel, Unity, ARCore

Tools: GitHub, Figma, Visual Studio, Docker (basic)

Databases: MySQL, Supabase, RDBMS Design

Other: Project Management, Analytical Thinking, Problem Solving, Technical Writing

PROJECTS

Weighted KNN Food Recommender · *KNN, FastAPI, Docker · Production System* 2026

- Designed Weighted Euclidean Distance KNN algorithm to recommend canteen food based on user preference vectors
- Deployed as FastAPI microservice in Docker container, integrated with NFC-based self-service ordering kiosk (Raspberry Pi)
- System handled orders for UC Makassar campus canteen as part of IoT + ML course capstone project

YOLOv11 Thermal Rat Detection System · *Computer Vision · Python, Ultralytics, OpenCV* 2025

- Trained YOLOv11 object detection model on thermal camera footage to detect rats in real-time across 3 location zones
- Built end-to-end inference pipeline from data labeling (Roboflow) to model evaluation, achieving mAP > 85%
- Reduced manual inspection effort by enabling automated 24/7 surveillance with frame-level detection output

Vaccine Demand Forecasting · *Apache Spark, scikit-learn, Random Forest · Published Research* 2025

- Built distributed ML pipeline on PySpark to predict regional vaccine demand across 34 provinces in Indonesia
- Applied Random Forest regression with feature engineering on multi-year epidemiological and demographic datasets
- Findings published in peer-reviewed journal; pipeline benchmarked against linear baseline with measurable RMSE improvement

TPOT AutoML — Kaggle Competition · *AutoML, scikit-learn, Python* 2024

- Applied TPOT genetic-algorithm AutoML to optimize full ML pipeline (preprocessing + model selection) for Kaggle dataset
- Evaluated and compared 5+ model configurations; final pipeline selected via cross-validated performance metrics

EDUCATION

Bachelor of Informatics (Specializing in Artificial Intelligence) · *Ciputra University, Makassar* 2023 – Present

- GPA: 3.90 / 4.00

- Relevant courses: Machine Learning, Big Data, Computer Vision, REST API, Mobile App Development, UI/UX Design, Database Design

EXPERIENCE

Secretary · *Informatics Student Union UC Makassar 24/25* Sep 2024 – Jul 2025

- Managed documentation and coordination for 3 student-led tech events including DevForge Hackathon and NPLC national competition, collectively serving 400+ participants
- Provided cross-divisional oversight and advisory support to Creative & Publication, Event, and Inventory divisions — monitoring progress, flagging blockers, and giving input to keep operations on track
- Contributed to structured reporting and post-event documentation for large-scale student competitions, seminars, and exhibitions

Coordinator, Publication & Documentation Dept. · *Orientation Week UC Makassar 2025* Dec 2024 – Sep 2025

- Led Publication & Documentation Division of 21 members + 3 coordinators (including self) within a 150+ person organizing committee
- Planned division task schedule, tracked execution, and coordinated with other division heads to ensure smooth overall execution
- Coordinated cross-functional teams across 4 divisions and handled administrative workflows for smooth event execution
- Contributed to visual design direction and event branding for 300+ attendee orientation program

ACHIEVEMENTS

2nd Place — UI/UX Design Competition · *Politeknik Negeri Bali* Oct 2025

- Designed user-centered interface with strong usability principles; placed 2nd among competing university teams

Regional Finalist — Hackathon Binar × Dispora S8 · *Makassar* Nov 2023

- Built AR-based tourism app using Unity + ARCore in 24-hour hackathon; selected as regional-level finalist from 15+ teams